

Q.22 Explain the various types of I/O modules used in PLC. (CO1)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw and explain memory organization in PLC with a diagram. (CO1)

Q.24 Write and explain ladder program for motor sequence control. (CO3)

Q.25 Explain serial port operation and timer programming in 8051 with examples. (CO5)

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No. of Printed Pages : 4
Roll No.

220943

4th Sem / Electrical

Subject : PLC and Microcontrollers

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which memory stores the user program in a PLC? (CO1)

- a) RAM b) ROM
- c) EEPROM d) Cache

Q.2 Which PLC block is responsible for logic execution? (CO1)

- a) Memory b) Power supply
- c) CPU d) I/O module

Q.3 What is the full form of ROM? (CO2)

- a) Read Only Memory
- b) Random Output Memory
- c) Real-time Output Mode
- d) None

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Q.4 Which register is used for serial communication in 8051? (CO5)

- a) TMOD b) PCON
- c) SBUF d) DPTR

Q.5 Traffic light control system is an example of: (CO4)

- a) Sequential control b) Batch control
- c) Continuous control d) None

Q.6 8051 microcontroller has how many 16-bit timers? (CO5)

- a) 1 b) 2
- c) 4 d) 8

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 CPU stands for _____. (CO1)

Q.8 _____ module is used to connect analog devices to PLC. (CO2)

Q.9 8051 has _____ pins in total. (CO4)

Q.10 The instruction used to move data in 8051 is _____. (CO5)

Q.11 Define Interrupt. (CO5)

Q.12 The _____ block in PLC executes logic and arithmetic functions. (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 List and explain any four advantages of PLC over conventional systems. (CO1)

Q.14 Differentiate between analog and discrete I/O modules. (CO1)

Q.15 Write a ladder diagram for ON-delay and OFF-delay timer. (CO2)

Q.16 Describe the use of arithmetic instructions in PLC programming. (CO2)

Q.17 Explain SFR. (CO4)

Q.18 Explain reset and clock circuit of 8051. (CO4)

Q.19 Write short notes on : (CO2)

- a) PSW Register
- b) SP register

Q.20 What are addressing modes in 8051? Give examples. (CO5)

Q.21 Explain the interfacing of keypad with 8051. (CO5)